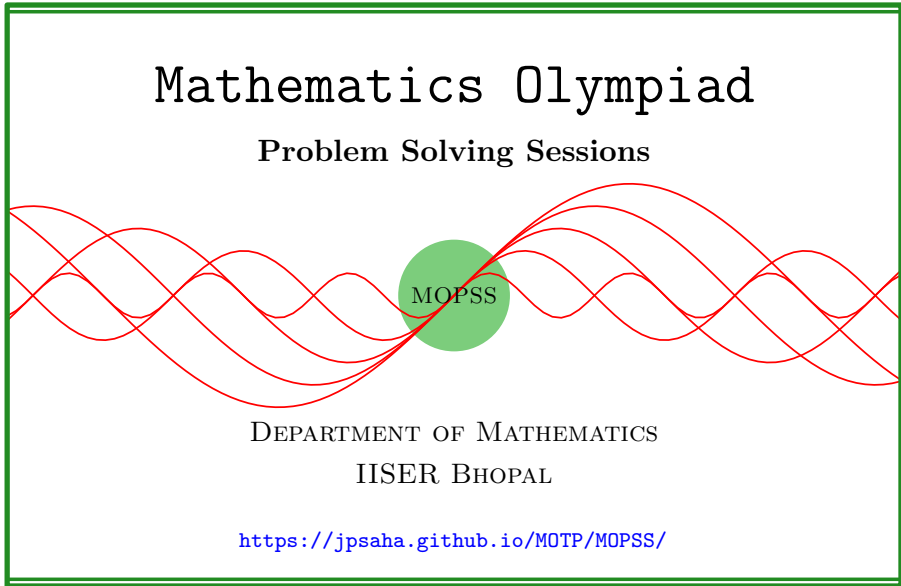


Orders

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1	Example (Tournament of Towns, India RMO 2014a P3) . . .	2
1.2	Exercise (Mathematical Ashes 2011 P2)	2

§1 Orders

Let p be a prime, and a be an integer, not divisible by p . The *order of a modulo p* , denoted by $\text{ord}_p(a)$, is defined to be the smallest positive integer such that $a^{\text{ord}_p(a)} \equiv 1 \pmod{p}$.

Example 1.1 (Tournament of Towns, India RMO 2014a P3). [Tao06, Problem 2.2] [AE11, Problem 3.81] Suppose for some positive integers r and s , 2^r is obtained by permuting the digits of 2^s in decimal expansion and $2^r, 2^s$ have same number of digits. Prove that $r = s$.

Solution 1. Since a positive integer is congruent to the sum of its digits modulo 9, it follows that 2^r and 2^s are congruent modulo 9.

Let us consider the case that $r < s$. Note that 9 divides $2^{s-r} - 1$. Since the order of 2 modulo 9 is equal to 6, it follows that 6 divides $s - r$, and hence $2^s \geq 64 \cdot 2^r$, which is impossible. This shows that $r \geq s$ holds. Similarly, it also follows that $s \geq r$ holds. This proves that $s = r$, as required. ■

Exercise 1.2 (Mathematical Ashes 2011 P2). Find all pairs (m, n) of non-negative integers for which

$$m^2 + 2 \cdot 3^n = m(2^{n+1} - 1).$$

Walkthrough —

- (a) Let m, n be nonnegative integers satisfying the given equation. Considering the roots of $x^2 - x(2^{n+1} - 1) + 2 \cdot 3^n$, it follows that

$$3^k + 2 \cdot 3^\ell = 2^{n+1} - 1$$

holds, for some nonnegative integers k, ℓ satisfying $k + \ell = n$.

- (b) Note that

$$3^k < 2^{n+1} < 9^{(n+1)/3}$$

holds, implying $k < 2(n+1)/3$. Also note that

$$2 \cdot 3^\ell < 2^{n+1} < 2 \cdot 3^{2n/3}$$

holds, implying $\ell < 2n/3$. Using $k + \ell = n$, show that

$$k > \frac{n-2}{3}, \ell > \frac{n-2}{3}.$$

(c) Let us consider the case^a that $n \geq 6$. Note that $m := \min\{k, \ell\} \geq 2$ holds.

(i) Note that 9 divides $2^{n+1} - 1$, and show that 6 divides $n + 1$. Writing $n + 1 = 6j$ yields

$$2^{n+1} - 1 = (4^j - 1)(4^{2j} + 4^j + 1) = (2^j - 1)(2^j + 1)((4^j - 1)^2 + 3 \cdot 4^j).$$

(ii) Noting that $(4^j - 1)^2 + 3 \cdot 4^j$ is divisible by 3, but not by 9, and that the integers $2^j - 1, 2^j + 1$ are coprime, conclude that 3^{m-1} divides one of $2^j - 1, 2^j + 1$.

(iii) Prove that

$$3^{m-1} \leq 2^j + 1 \leq 3^j = 3^{\frac{n+1}{6}},$$

implying

$$m - 1 \leq \frac{n + 1}{6}.$$

(iv) Conclude that

$$\frac{n - 2}{3} - 1 < m - 1 \leq \frac{n + 1}{6}.$$

holds.

(v) This yields $n < 11$, contradicting $n \geq 6$ and 6 divides $n + 1$.

(d) It remains to consider the case $n \leq 5$.

^aIt also suffices to assume that $n \geq 5$ holds to obtain $m \geq 2$.

Solution 2. Let m, n be non-negative integers satisfying satisfying the given equation. Since the quadratic polynomial $x^2 - x(2^{n+1} - 1) + 2 \cdot 3^n$ has one integer root, it follows that its roots are integers, and are equal to $3^k, 2 \cdot 3^\ell$ for some nonnegative integers k, ℓ satisfying $k + \ell = n$. This gives

$$3^k + 2 \cdot 3^\ell = 2^{n+1} - 1.$$

Using $2^{n+1} < 3^{\frac{2(n+1)}{3}}$, we have

$$3^k < 3^{\frac{2(n+1)}{3}}$$

and

$$2 \cdot 3^\ell < 2^{n+1} < 2 \cdot 3^{\frac{2(n+1)}{3}}$$

hold, which shows that

$$k < \frac{2(n+1)}{3}, \ell < \frac{2(n+1)}{3}.$$

Since $k + \ell = n$, we obtain

$$k > \frac{n-2}{3}, \ell > \frac{n-2}{3}.$$

Let us consider the case that $n \geq 5$. Let m denote the minimum of the integers k, ℓ . Note that $m \geq 2$ holds. Since 9 divides $2^{n+1} - 1$, it follows that 6 divides $n + 1$. Write $n + 1 = 6j$ for some positive integer j . This gives

$$2^{n+1} - 1 = (4^j - 1)(4^{2j} + 4^j + 1) = (2^j - 1)(2^j + 1)((4^j - 1)^2 + 3 \cdot 4^j).$$

Note that $(4^j - 1)^2 + 3 \cdot 4^j$ is divisible by 3, but not by 9, and that the integers $2^j - 1, 2^j + 1$ are coprime being consecutive odd integers. This shows that 3^{m-1} divides one of $2^j - 1, 2^j + 1$. It follows that

$$3^{m-1} \leq 2^j + 1 \leq 3^j = 3^{\frac{n+1}{6}},$$

which yields

$$m - 1 \leq \frac{n + 1}{6}.$$

Combining this with

$$\frac{n - 2}{3} < m,$$

we obtain

$$\frac{n - 2}{3} - 1 < \frac{n + 1}{6},$$

which gives $n < 11$. Since $n + 1$ is divisible by 6 and $n \geq 5$, it follows that $n = 5$ holds. Therefore, it remains to consider the case that $n \leq 5$.

Note that the discriminant of the quadratic polynomial $x^2 - x(2^{n+1} - 1) + 2 \cdot 3^n$ is equal to

$$(2^{n+1} - 1)^2 - 4 \cdot 2 \cdot 3^n,$$

which is a perfect square. This implies that n is not equal to any of 0, 1, 2, 4. Consequently, for any nonnegative integers m, n satisfying the given equation, the integer n is equal to either 3 or 5. Checking these cases, we obtain the solutions

$$(6, 3), (9, 3), (9, 5), (54, 5),$$

which are precisely all the solutions of the given equation over the nonnegative integers. ■

References

- [AE11] TITU ANDREESCU and BOGDAN ENESCU. *Mathematical Olympiad treasures*. Second. Birkhäuser/Springer, New York, 2011, pp. viii+253. ISBN: 978-0-8176-8252-1; 978-0-8176-8253-8 (cited p. 2)
- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)
- [Tao06] TERENCE TAO. *Solving mathematical problems*. A personal perspective. Oxford University Press, Oxford, 2006, pp. xii+103. ISBN: 978-0-19-920560-8; 0-19-920560-4 (cited p. 2)